Year 11 Higher

# REVISION PACK

## Number 8

**Name:** _____

**Class:** _____

## Number 8 · Factors, multiples and primes (Lessons 1.2, 1.3)

**Worked example: prime factors in index form**

Write 252 as a product of its prime factors.
Give your answer in index form.

| | | |
|---|---|---|
| **Step 1** | Divide by 2 | $252 = 2 \times 126$ |
| **Step 2** | Divide by 2 | $126 = 2 \times 63$ |
| **Step 3** | Divide by 3 | $63 = 3 \times 21$ |
| **Step 4** | Divide by 3 | $21 = 3 \times 7$ |
| **Step 5** | Answer | $2^2 \times 3^2 \times 7$ |

### Practice: prime factors in index form

**1.1** Write 120 as a product of its prime factors. Give your answer in index form.

**1.2** Write 504 as a product of its prime factors. Give your answer in index form.

**1.3** Write 600 as a product of its prime factors. Give your answer in index form.

**1.4** Write 280 as a product of its prime factors. Give your answer in index form.

**1.5** Write 450 as a product of its prime factors. Give your answer in index form.

**1.6** Write 792 as a product of its prime factors. Give your answer in index form.

## Number 8 · Factors, multiples and primes (Lessons 1.2, 1.3)

**Worked example: HCF and LCM**

Find

(a) the highest common factor (HCF) of 60 and 72 (b) the lowest common multiple (LCM) of 60 and 72

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Prime factors of 60 | $60 = 2^2 \times 3 \times 5$ |
| | **Step 2** | Prime factors of 72 | $72 = 2^3 \times 3^2$ |
| | **Step 3** | HCF: primes in both | $2^2 \times 3 = 12$ |
| **(b)** | **Step 1** | LCM: highest powers | $2^3 \times 3^2 \times 5 = 360$ |

### Practice: HCF and LCM

**1.7** Find

(a) the highest common factor (HCF) of 90 and 150 (b) the lowest common multiple (LCM) of 90 and 150

**1.8** Find

(a) the highest common factor (HCF) of 48 and 180 (b) the lowest common multiple (LCM) of 48 and 180

**1.9** Find

(a) the highest common factor (HCF) of 72 and 108 (b) the lowest common multiple (LCM) of 72 and 108

**1.10** Find

(a) the highest common factor (HCF) of 56 and 84 (b) the lowest common multiple (LCM) of 56 and 84

**1.11** Find

(a) the highest common factor (HCF) of 126 and 168 (b) the lowest common multiple (LCM) of 126 and 168

**1.12** Find

(a) the highest common factor (HCF) of 60 and 96 (b) the lowest common multiple (LCM) of 60 and 96

## Number 8 · Factors, multiples and primes (Lessons 1.2, 1.3)

**Worked example: next time together**

Bus A leaves the bus station every 15 minutes. Bus B leaves the bus station every 20 minutes. Both buses leave the bus station at 9:00 am. When will both buses next leave the bus station at the same time?

| | | |
|---|---|---|
| **Step 1** | LCM of 15 and 20 | $60$ minutes |
| **Step 2** | Add to 9:00 am | $9:00 + 60$ minutes |
| **Step 3** | Answer | $10:00$ am |

### Practice: next time together

**1.13** Bus A leaves the bus station every 16 minutes. Bus B leaves the bus station every 24 minutes. Both buses leave the bus station at 7:00 am. When will both buses next leave the bus station at the same time?

**1.14** Train A leaves the station every 10 minutes. Train B leaves the station every 25 minutes. Both trains leave the station at 8:30 am. When will both trains next leave the station at the same time?

**1.15** Bus A leaves the bus station every 14 minutes. Bus B leaves the bus station every 21 minutes. Both buses leave the bus station at 6:00 am. When will both buses next leave the bus station at the same time?

**1.16** Train A leaves the station every 20 minutes. Train B leaves the station every 30 minutes. Both trains leave the station at 9:15 am. When will both trains next leave the station at the same time?

**1.17** Bus A leaves the bus station every 18 minutes. Bus B leaves the bus station every 24 minutes. Both buses leave the bus station at 7:30 am. When will both buses next leave the bus station at the same time?

**1.18** Bus A leaves the bus station every 12 minutes. Bus B leaves the bus station every 15 minutes. Both buses leave the bus station at 10:00 am. When will both buses next leave the bus station at the same time?

## Number 8 · Standard form (Lesson 1.1)

**Worked example: calculating in standard form**

Work out. Give your answers in standard form.

(a) $(2.5 \times 10^3) \times (6 \times 10^{-5})$ (b) $(3 \times 10^4) + (8.2 \times 10^3)$

| | | | |
|---|---|---|---|
| (a) | Step 1 | Multiply the numbers | $2.5 \times 6 = 15$ |
| | Step 2 | Add the powers | $10^{3+(-5)} = 10^{-2}$ |
| | Step 3 | Answer | $1.5 \times 10^{-1}$ |
| (b) | Step 1 | Ordinary numbers | $30\,000 + 8200 = 38\,200$ |
| | Step 2 | Answer | $3.82 \times 10^4$ |

### Practice: calculating in standard form

**2.1** Work out. Give your answers in standard form.

(a) $(4 \times 10^3) \times (2.5 \times 10^{-6})$ (b) $(5 \times 10^4) + (3.2 \times 10^3)$

**2.2** Work out. Give your answers in standard form.

(a) $(1.6 \times 10^5) \times (5 \times 10^{-2})$ (b) $(7 \times 10^5) + (6.5 \times 10^4)$

**2.3** Work out. Give your answers in standard form.

(a) $(2.4 \times 10^{-3}) \times (3 \times 10^8)$ (b) $(2 \times 10^6) + (8.5 \times 10^5)$

**2.4** Work out. Give your answers in standard form.

(a) $(6 \times 10^4) \times (1.5 \times 10^{-9})$ (b) $(4.1 \times 10^3) + (9 \times 10^2)$

**2.5** Work out. Give your answers in standard form.

(a) $(3.5 \times 10^2) \times (4 \times 10^{-5})$ (b) $(8 \times 10^4) + (2.4 \times 10^3)$

**2.6** Work out. Give your answers in standard form.

(a) $(2.5 \times 10^{-4}) \times (8 \times 10^7)$ (b) $(1.2 \times 10^5) + (6 \times 10^4)$

## Number 8 · Standard form (Lesson 1.1)

**Worked example: writing numbers in standard form**

This question is about standard form.

(a) Write 0.000507 in standard form. (b) Write $2.4 \times 10^{-4}$ as an ordinary number.

| | | | |
|---|---|---|---|
| (a) | Step 1 | Point after the 5 | $5.07$ |
| | Step 2 | Small number: negative power | $5.07 \times 10^{-4}$ |
| (b) | Step 1 | Divide by 10 000 | $2.4 \div 10\,000 = 0.00024$ |

### Practice: writing numbers in standard form

**2.7** This question is about standard form.

(a) Write 0.00072 in standard form.

(b) Write $3.6 \times 10^{-4}$ as an ordinary number.

**2.8** This question is about standard form.

(a) Write 0.0000415 in standard form.

(b) Write $5.2 \times 10^{-3}$ as an ordinary number.

**2.9** This question is about standard form.

(a) Write 0.00908 in standard form.

(b) Write $1.75 \times 10^{-5}$ as an ordinary number.

**2.10** This question is about standard form.

(a) Write 0.000063 in standard form.

(b) Write $8.04 \times 10^{-2}$ as an ordinary number.

**2.11** This question is about standard form.

(a) Write 0.0123 in standard form.

(b) Write $4.9 \times 10^{-6}$ as an ordinary number.

**2.12** This question is about standard form.

(a) Write 0.000000208 in standard form.

(b) Write $6.3 \times 10^{-3}$ as an ordinary number.

## Number 8 · Ratio (Lesson 1.4)

**Worked example: sharing in a ratio**

Share 240 KD in the ratio 1 : 3 : 4

| | | |
|---|---|---|
| **Step 1** | Add the parts | $1 + 3 + 4 = 8$ |
| **Step 2** | One part | $240 \div 8 = 30$ |
| **Step 3** | 1 part | $1 \times 30 = 30$ |
| **Step 4** | 3 parts | $3 \times 30 = 90$ |
| **Step 5** | 4 parts | $4 \times 30 = 120$ |
| **Step 6** | Answer | $30\ \text{KD},\ 90\ \text{KD},\ 120\ \text{KD}$ |

### Practice: sharing in a ratio

**3.1** Share 300 KD in the ratio 2 : 3 : 5

**3.2** Share 420 KD in the ratio 1 : 2 : 4

**3.3** Share 540 KD in the ratio 2 : 3 : 4

**3.4** Share 600 KD in the ratio 3 : 4 : 5

**3.5** Share 330 KD in the ratio 1 : 4 : 6

**3.6** Share 720 KD in the ratio 1 : 3 : 5

## Number 8 · Ratio (Lesson 1.4)

**Worked example: ratio and difference**

The ratio of boys to girls in a club is 3 : 5. There are 14 more girls than boys. How many boys and girls are there altogether?

| | | |
|---|---|---|
| **Step 1** | Difference in parts | $5 - 3 = 2$ |
| **Step 2** | One part | $14 \div 2 = 7$ |
| **Step 3** | Total parts | $3 + 5 = 8$ |
| **Step 4** | Answer | $8 \times 7 = 56$ |

### Practice: ratio and difference

**3.7** The ratio of boys to girls in a club is 2 : 5. There are 18 more girls than boys. How many boys and girls are there altogether?

**3.8** The ratio of cats to dogs in a shelter is 3 : 7. There are 20 more dogs than cats. How many cats and dogs are there altogether?

**3.9** The ratio of red cars to blue cars in a car park is 4 : 7. There are 15 more blue cars than red cars. How many red cars and blue cars are there altogether?

**3.10** The ratio of adults to children in a cinema is 5 : 8. There are 12 more children than adults. How many adults and children are there altogether?

**3.11** The ratio of teachers to students in a school trip is 1 : 4. There are 27 more students than teachers. How many teachers and students are there altogether?

**3.12** The ratio of boys to girls in a class is 3 : 4. There are 9 more girls than boys. How many boys and girls are there altogether?